A Physics-Conditioned 3D U-Net for Proton Dose Prediction on CT

Jyothi Sree Masina*

Independent submission, DoseRAD2026 Grand Challenge

Abstract. This work extends the beam-conditioned 3D U-Net developed for our photon dose-prediction submissions to proton pencil-beam scanning (PBS), which delivers dose as a superposition of narrow, mono-energetic beamlets rather than a broad MLC-shaped field and so admits no direct analogue of an aperture mask. In place of ray tracing, we built an analytic beamlet encoder from first-principles physics – range from the Bragg-Kleeman relation, range straggling from Bortfeld's mono-energetic approximation, lateral spread from the Highland multiple-Coulomb-scattering formula – feeding the same 3D U-Net (4.8M parameters, unchanged) at a finer, patient-specific $176 \times 512 \times 512$ grid. Proton's larger grid immediately broke a memory assumption that had been fine at the photon tasks' scale (a per-patient coordinate-grid cache that ballooned to roughly 37 GB across the training set), which we traced and fixed by sharing one origin-free coordinate grid across patients instead. Our real submission to the Preliminary Testing Phase – an epoch-6 checkpoint trained without any tissue-density correction to beam range – was scored with IDD curve distance disproportionately worse (roughly $320\times$ the leading entry) than overall Beam MAE ($\sim 9\times$), which points fairly clearly at systematic Bragg-peak mispositioning. Chasing that down after the fact, we found and fixed a real bug in a water-equivalent-path-length correction we had implemented but never actually validated end-to-end – a bad fallback for beamlets exiting near the patient surface, off by as much as 121 mm on one beamlet we checked – and fine-tuned from the submitted checkpoint with the fix applied. Validation loss never beat the original checkpoint across three fine-tune epochs, so that fine-tune wasn't deployed. Digging further into why, we found something more concerning: the submitted checkpoint's training run had actually finished before the correction was even wired into the training loop, meaning the deployed model was scored on inputs it was never trained to see. That's reported here as a real, unresolved limitation of the current submission.

Keywords: dose prediction, deep learning, radiotherapy, 3D U-Net, Monte Carlo surrogate, proton therapy, pencil beam scanning

* Corresponding author. Email: `jyothisreedurgamasina@gmail.com`

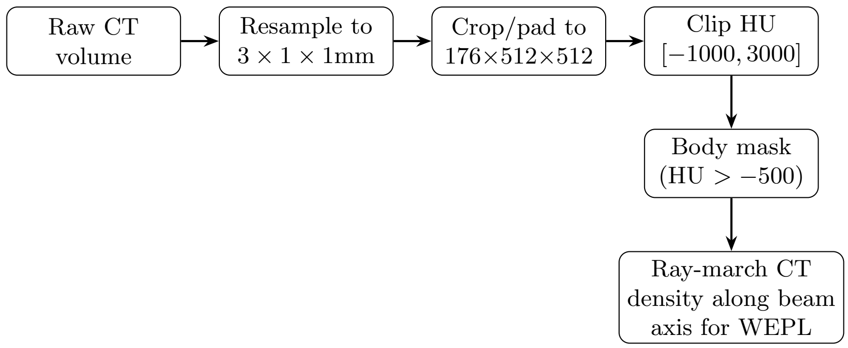

Fig. 1. Proton preprocessing pipeline. The finer grid and the water-equivalent-path-length step (Section 2.2) are what actually differ from the photon pipeline.

1 Introduction

Proton pencil-beam scanning delivers dose through many narrow, mono-energetic beamlets rather than a single broad field, exploiting the Bragg peak's sharp, finite-range falloff for conformality photon therapy can't match. This work poses proton dose prediction on CT as the same underlying problem as our photon submissions – predict Geant4 [1] Monte Carlo ground truth from patient imaging and beam parameters – but the beam description is physically different: a plan is a set of beams, each with multiple rays (fixed source and target positions), each ray with multiple beamlets at discrete energies, rather than control points along a continuous gantry rotation.

What follows covers how we extended the shared architecture to that physics, a memory bug proton's larger voxel grid exposed, our actual leaderboard result, and a fairly involved post-submission investigation into that result's biggest weakness – including a limitation we found but didn't manage to fix before this report was due.

2 Methods

2.1 Data

Each patient provides a planning CT and a set of proton beams structured as beams $\rightarrow$ rays $\rightarrow$ beamlets, each beamlet defined by its energy and its ray's source/target position. Ground truth is provided per beamlet, on a substantially finer, patient-specific grid ($1 \times 1 \times 3$ mm spacing, in-plane extents up to roughly 500 voxels) than the photon submissions' fixed 256^3 grid. The full training set spans about 77 patients and 81,000 beamlets (∼423 GB), which we downloaded and cross-checked against real patient geometry – grid spacing, extent, beam JSON schema – on five patients before training began.

Each CT volume is resampled to a fixed $176 \times 512 \times 512$ shape at $3 \times 1 \times 1$ mm spacing – chosen to match the finer proton dose grid rather than reusing

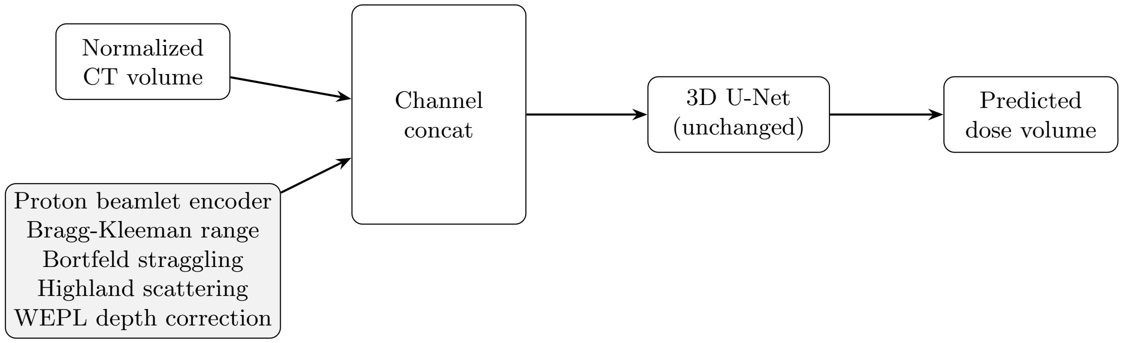

Fig. 2. The network and conditioning mechanism are unchanged from the photon work – only the encoder producing the beam mask differs, now built from proton physics rather than MLC ray tracing.

the photon submissions' coarser resolution – with the same HU clipping and −500 HU body-mask threshold as the CT photon work. Dose targets are scaled by a constant of 1000 (different from the photon submissions' 10,000, since proton dose magnitudes are on a different scale). The same intensity-only augmentation and patient-level 90/10 split (8 of 75 patients held out) apply here too. No data beyond the challenge's official proton release was used.

2.2 Model

The 3D U-Net [2] itself – 5-level encoder/decoder, 16–32–64–128–256 channels, 2 residual units per block, instance normalization, 4,806,398 parameters, default initialization, nothing pretrained, implemented with MONAI's [5] `UNet` – and the early channel-concatenation conditioning are unchanged from the photon submissions. What's different is the beam encoder producing that second input channel.

Beamlet Encoding. A beamlet's geometry – source, target, energy – has no MLC-aperture analogue, so we built the encoder from first-principles physics instead of adapting the photon ray tracer. For energy E (MeV), we compute range via the Bragg-Kleeman relation, $R\,[\mathrm{mm}] = 10 \cdot 0.0022 \cdot E^{1.77}$; range straggling via Bortfeld's [4] mono-energetic approximation, $\sigma_R\,[\mathrm{mm}] = 10 \cdot 0.012 \cdot R\,[\mathrm{cm}]^{0.935}$; and lateral spread via the Highland [3] formula, evaluated in water ($X_0 = 360.8\,\mathrm{mm}$) with relativistic kinematics for a proton (rest mass 938.272 MeV). Each voxel's signed depth along the beamlet axis and lateral distance from it feed a single Gaussian in depth (centered at entry-to-body depth plus range, width σ_R) and a single Gaussian laterally, matching the single-Gaussian pencil-beam model the challenge itself uses as reference. Entry-to-body depth is found by marching the depth axis against the body mask, since a beamlet's source sits in air well outside the patient rather than at the body surface – an early version measured range from the source instead and produced a mask that was systematically wrong;

we only caught it by checking predicted peak-dose location against real beamlet dose files (3–17 mm accuracy across five beamlets we sampled) before trusting the encoder in training.

A Memory Bug Proton's Grid Exposed. Both encoders had originally been written to cache one full-volume coordinate-grid tensor per patient – a reasonable choice at the photon tasks' fixed 256^3 grid ($\sim$200 MB per tensor, $\sim$13.4 GB across 67 training patients), but not at proton's larger grid ($\sim$553 MB per tensor, $\sim$37 GB across the same patient count). Training crashed with a CUDA out-of-memory error partway through the first epoch, once enough distinct shuffled patients had been visited to fill the cache. The actual fix, once we did the memory arithmetic, was straightforward: the coordinate grid itself doesn't depend on which patient it's for, only on volume shape and spacing, both fixed by preprocessing – what varies per patient is just the isocenter offset, a tiny $(3,)$ tensor. Refactoring both encoders to share one origin-free grid and apply each patient's offset separately collapsed the cache to a single shared entry and eliminated the OOM, with identical beamlet peak-distance numbers before and after confirming nothing about the model's actual output had changed.

Water-Equivalent-Path-Length Correction. The encoder above assumes the beamlet's whole in-patient path is water when deciding where along the axis the Bragg peak falls, ignoring real tissue density from the CT. We implemented a correction that samples CT-derived stopping power along the beamlet axis and finds the water-equivalent depth at which the beamlet's range runs out, falling back to the water-only estimate when a beamlet's path doesn't accumulate enough water-equivalent depth within the sampling window to reach its nominal range – which happens for beamlets exiting near or through the patient surface.

2.3 Training

Optimizer, schedule, loss composition, and regularization are unchanged from the photon submissions: AdamW at 2×10^{-4}, weight decay 1×10^{-5}, cosine annealing, mixed precision, gradient clipping at norm 1.0, batch size 1, the same finite-value guard on loss and gradients. The three-term loss ($w_m = 1.0$, $w_b = w_o = 0.2$, including the outside-body term from the photon leakage fix) carried over with no proton-specific change. The checkpoint behind our real Preliminary Testing Phase submission is epoch 6 of the original training run, chosen by validation loss; its exact relationship to the WEPL correction above is discussed in Section 3.

2.4 Evaluation

Same local protocol as the photon work – Masked MAE, IDD curve distance, Stratified Plan MAE, Gamma pass rate at 1%/1 mm, DVH score unavailable – extended to the proton beam/ray/beamlet index and encoder. Beyond local evaluation, this is the one submission of the four described across our reports with a real Grand Challenge Preliminary Testing Phase leaderboard result at

Table 1. Fine-tune validation loss by epoch, resumed from the submitted epoch-6 checkpoint. Baseline is that checkpoint's own validation loss.

Run	Epoch(s)	Validation total loss
Submitted checkpoint (no WEPL wiring)	6	0.0802
First fine-tune (buggy WEPL)	7–8	0.0974 → 0.1035
Second fine-tune (fixed WEPL)	7 / 8 / 9	0.0878 / 0.0888 / 0.0882

the time of writing, which we treat as the most authoritative accuracy signal available for it. We report inference runtime against the packaged container where we have it. As with the other submissions, we report point estimates without formal confidence intervals and didn't run any post-hoc explainability analysis.

3 Results

3.1 The leaderboard result

Our submission (2026-08-25) was scored on the Preliminary Testing Phase leaderboard. The breakdown across scored metrics pointed at a specific weakness rather than a generic accuracy gap: IDD curve distance was roughly 320× worse than the leading entry, against about 9× worse on overall Beam MAE. Since IDD curve distance specifically measures depth-dose-profile shape – in practice, Bragg-peak position – that gap is consistent with the water-only range assumption discussed above: the submitted checkpoint's training predates the WEPL correction entirely, a point we come back to in Section 3.3.

3.2 Chasing down and fixing the WEPL bug

After the submission above, we found a real bug in the WEPL correction's fallback path. For a beamlet whose in-patient path didn't accumulate enough water-equivalent depth to reach its nominal range within the sampling window, the fallback we'd written clamped to the last sampled index – a fixed jump of roughly 349 mm that had nothing to do with the beamlet's actual physics, and which was off by as much as 121 mm on one real beamlet we checked. We changed the fallback to the water-only estimate instead. A quick diagnostic afterward (mean WEPL error across a sample of beamlets) showed 6.5 mm mean error post-fix, beating the water-only baseline's 7.6 mm – and confirming the pre-fix, buggy version had actually been worse than no correction at all, at 8.2 mm mean error.

We fine-tuned from the submitted checkpoint with the fix applied, this time actually training with the WEPL-corrected input rather than just patching inference. A first attempt, run before we'd found the fallback bug, made things worse (Table 2, row 2) – both epochs landed above the submitted checkpoint's own 0.0802. After the bug fix, a second three-epoch fine-tune (row 3) improved a lot over the buggy attempt but plateaued around 0.088 across all three epochs, never getting back to the submitted checkpoint's validation loss, while training loss kept dropping – the same overfitting signature we'd already seen on the MR photon submission. We didn't deploy this fine-tune; the original submitted checkpoint is still our live entry.

3.3 A mismatch we found in our own live submission

While trying to understand why the fix didn't translate into a better fine-tune, we checked exactly when the submitted checkpoint's training run finished relative to when the WEPL correction actually got wired into the training loop's encoder call. The file timestamps told a clear story: the submitted checkpoint was written at 04:41 on the day of submission; the code change that started passing CT data into the encoder during training landed at 15:03 that same day – after the checkpoint's training had already finished. In other words, the submitted checkpoint's weights were trained entirely under the water-only assumption, never once seeing WEPL-corrected input during training, while the submission container's inference code does pass that corrected input at prediction time. That's a genuine mismatch between what the model was trained on and what it's being asked to handle at inference, sitting inside an already-scored submission, independent of (and probably compounding) the Bragg-peak weakness visible in the leaderboard breakdown itself. We haven't isolated how much each factor contributes individually.

4 Discussion

The lesson from this submission is that a beam encoder's physics correction has to be checked for training/inference consistency, not just geometric accuracy on its own. The WEPL fix's spot-check looked like a real improvement in isolation, and the encoder logic itself was correct once the fallback bug was fixed – neither of those facts guarantees a deployed checkpoint actually benefits, because the checkpoint's weights only ever encode whatever it was actually trained on. We only found this by checking file timestamps against a code-change timeline after an unexpected fine-tune result, not through any systematic pre-deployment check – which is really the most useful thing to take from this, independent of the specific proton physics involved.

I will be direct about how I feel about this particular submission: it is the one I am least comfortable with. Finding a genuine train/inference mismatch inside an entry that has already been scored is not a satisfying place to end a report, but reporting it honestly seemed like the only defensible option once I had confirmed it against the timestamps myself. If nothing else, it has taught me to check exactly this kind of thing before submitting, not after – a lesson that arrived one submission too late for this particular entry.

Limitations. The submitted, scored entry is known to have the mismatch described in Section 3.3; we haven't corrected and resubmitted it before this report's deadline, so the scored result reflects a submission we now know has this specific, verified defect. Neither fine-tune attempt recovered the submitted checkpoint's own validation loss, so we don't have a validated better alternative ready to deploy – retraining from initialization with the WEPL correction wired in from the start, rather than patched in via a short fine-tune, is unexplored and seems like the more promising next step, but we haven't attempted it given the time available. No per-beamlet inference latency was measured locally for this submission; the only evidence it clears the 1-second-per-beam budget is that the platform accepted and scored it, which enforces that limit server-side – we can't independently confirm the margin. Our local evaluation wasn't run to completion against this submission's full held-out set at the time of writing, so the leaderboard result in Section 3.1, not a locally computed table, is our primary accuracy evidence here, unlike the photon submissions.

5 Author Contributions

Jyothi Sree Masina: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review & editing, Visualization, Supervision, Project administration.

6 Other Information

Funding. This work received no external funding.

Availability. Source code (training pipeline, proton beamlet encoder, WEPL correction, evaluation harness, and Docker packaging scripts) is being made available in a dedicated public repository for this submission, consistent with the challenge's open-source requirement for winning teams.

Disclosure of Interests. The author has no competing interests to declare that are relevant to the content of this article.

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